

Department of Computer Science and Engineering
Islamic University of Technology (IUT)
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Laboratory Report

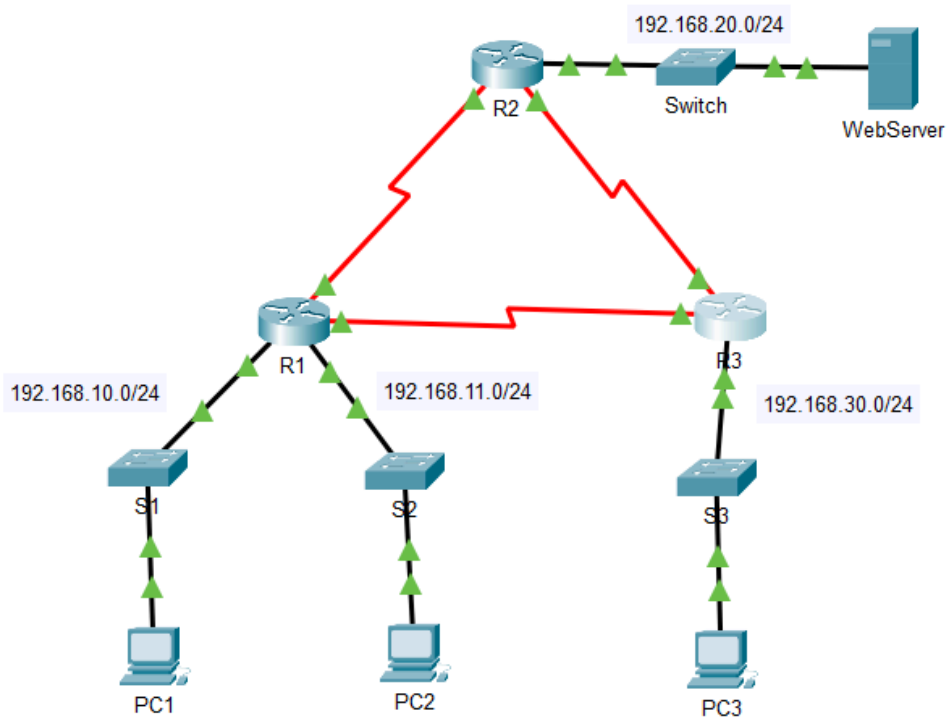
CSE 4512: Computer Networks Lab

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Section: B
Semester: 4th
Academic Year: 2022-2023

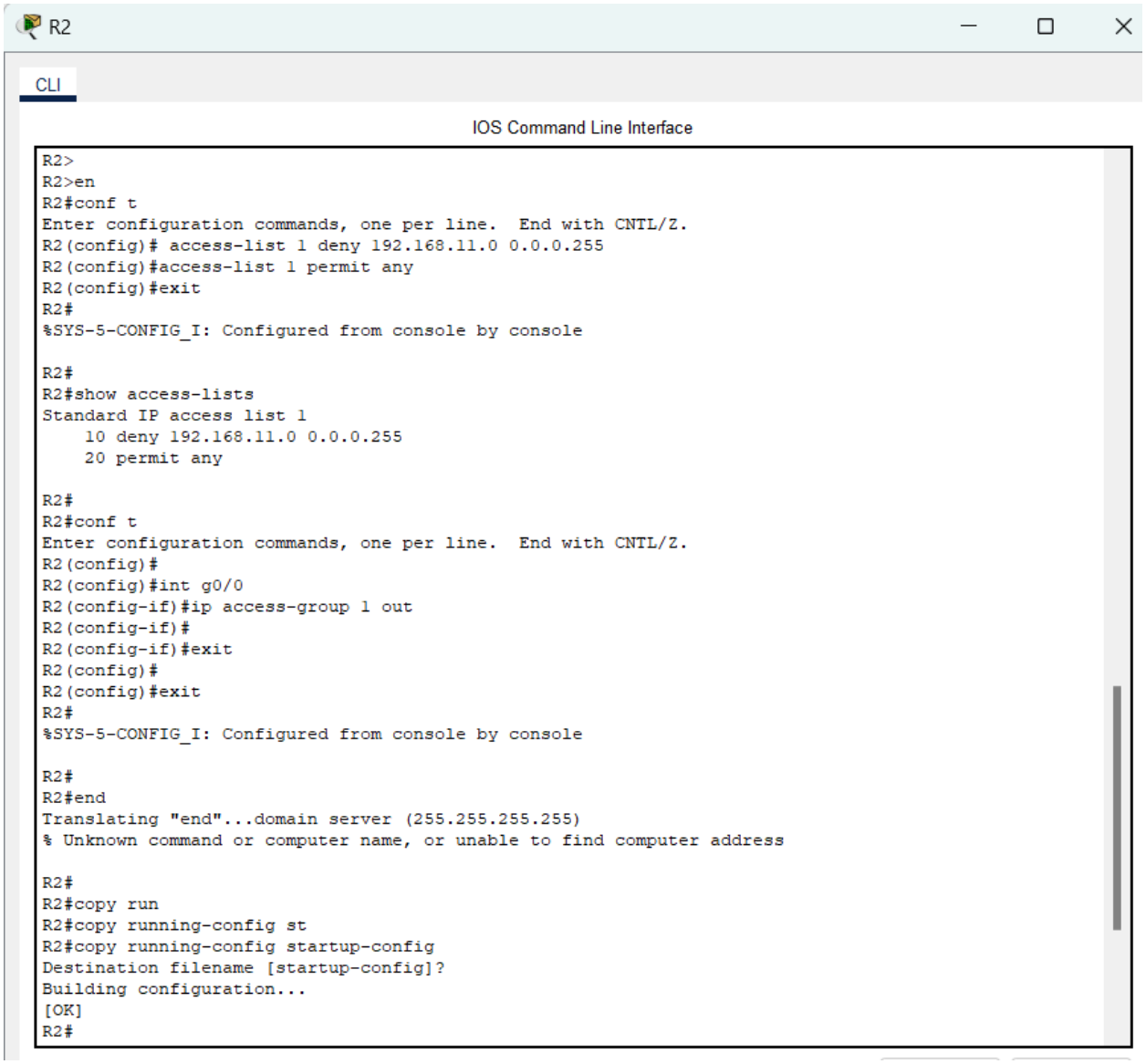
Date of Submission: 05/04/2024

Title: Configuring ACL and NAT in Cisco Devices

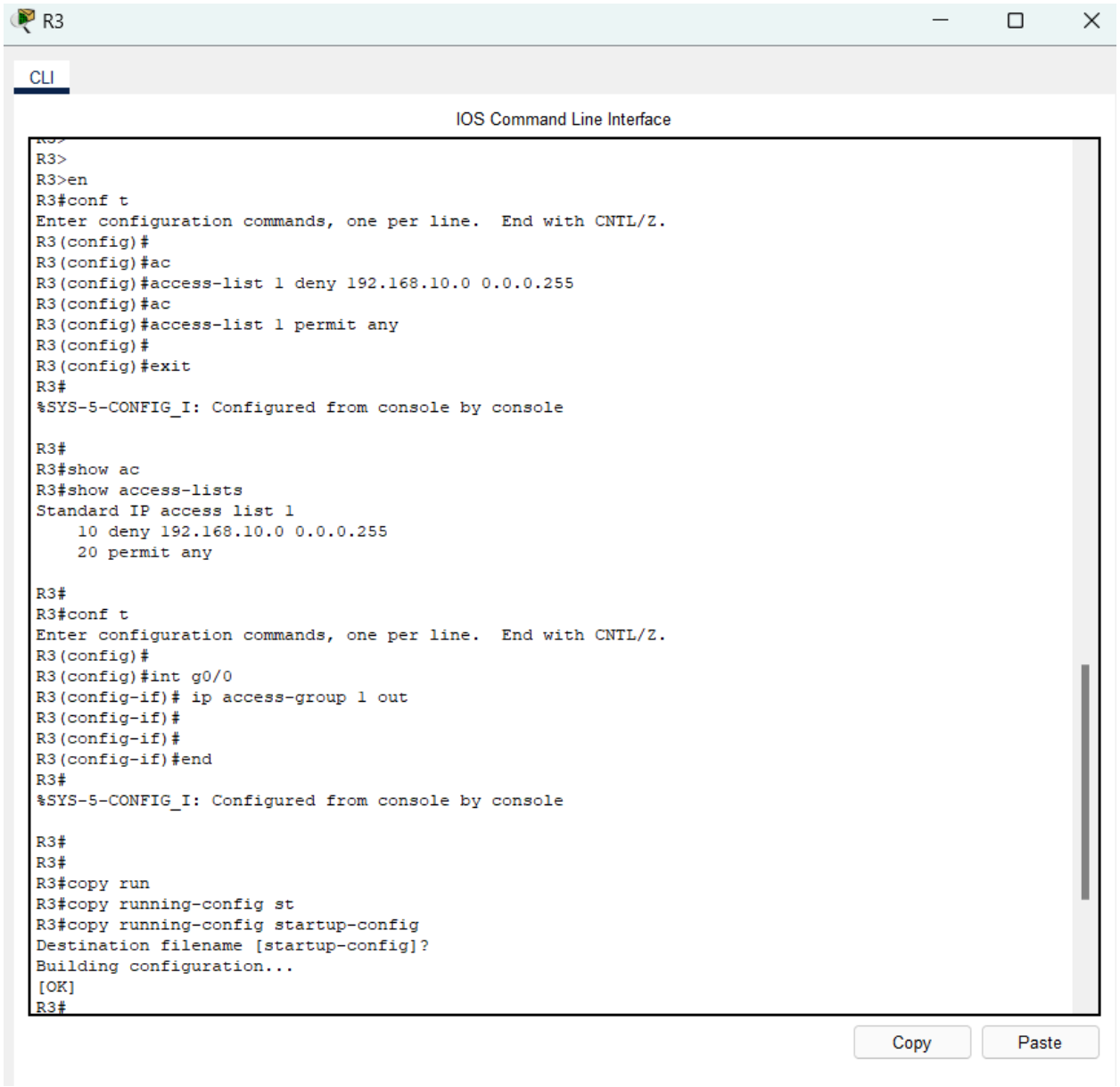
Diagram of the experiment(s):



Working Procedure:

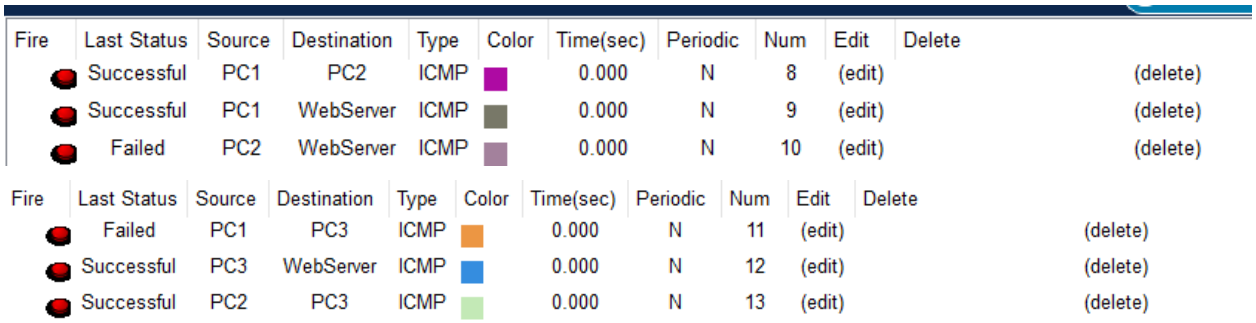



```
R2#
R2#
R2#
R2#show ac
R2#show access-lists
Standard IP access list 1
    10 deny 192.168.11.0 0.0.0.255 (1 match(es))
    20 permit any (2 match(es))
```




```
R3#
R3#
R3#show ac
R3#show access-lists
Standard IP access list 1
    10 deny 192.168.10.0 0.0.0.255 (1 match(es))
    20 permit any (2 match(es))
```

Observation:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC1	WebServer	ICMP		0.000	N	9	(edit)	(delete)
	Failed	PC2	WebServer	ICMP		0.000	N	10	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC1	PC3	ICMP		0.000	N	11	(edit)	(delete)
	Successful	PC3	WebServer	ICMP		0.000	N	12	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	13	(edit)	(delete)

Questions:

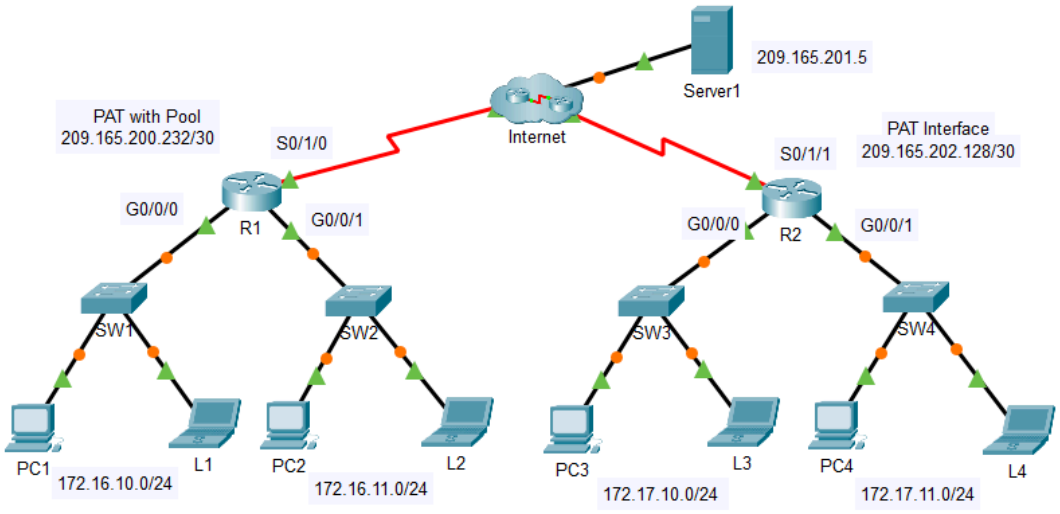
Task # 01:

1. The ping from 192.168.10.10 to 192.168.11.10 is successful or not? Explain.
   Ans: Successful. Because there weren't any restrictions or access-lists provided for the 'in' or 'out' for the network 192.168.11.0/24.

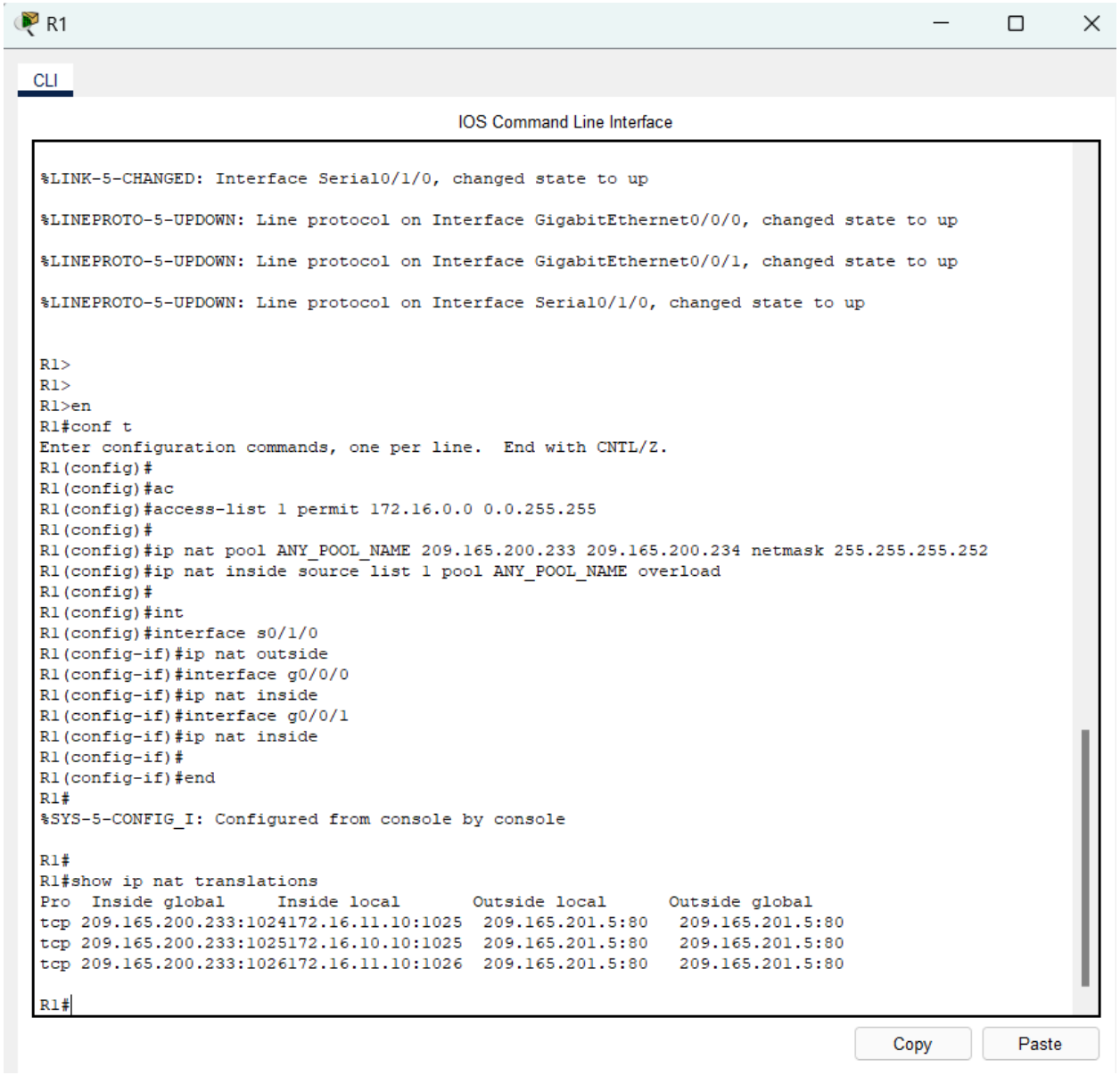
2. The ping from 192.168.10.10 to 192.168.20.254 is successful or not? Explain.
   Ans: Successful. Because the 192.168.10.0/24 network is allowed access to the WebServer on the 192.168.20.0/24 network.

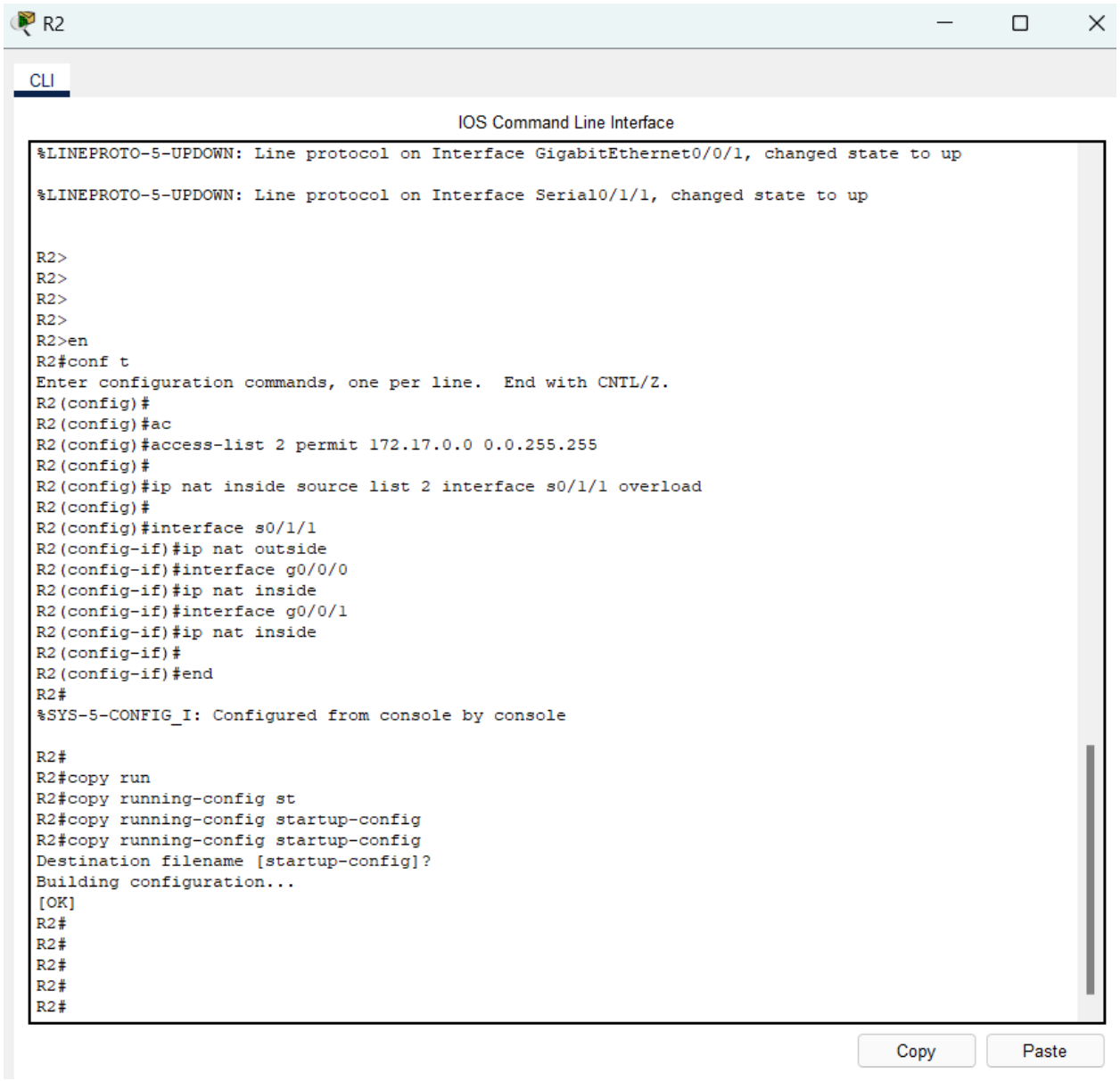
3. The ping from 192.168.11.10 to 192.168.20.254 failed or not? Explain.
   Ans: Failed. Because the 192.168.11.0/24 network is not allowed access to the WebServer on the 192.168.20.0/24 network.

Diagram of the experiment(s):

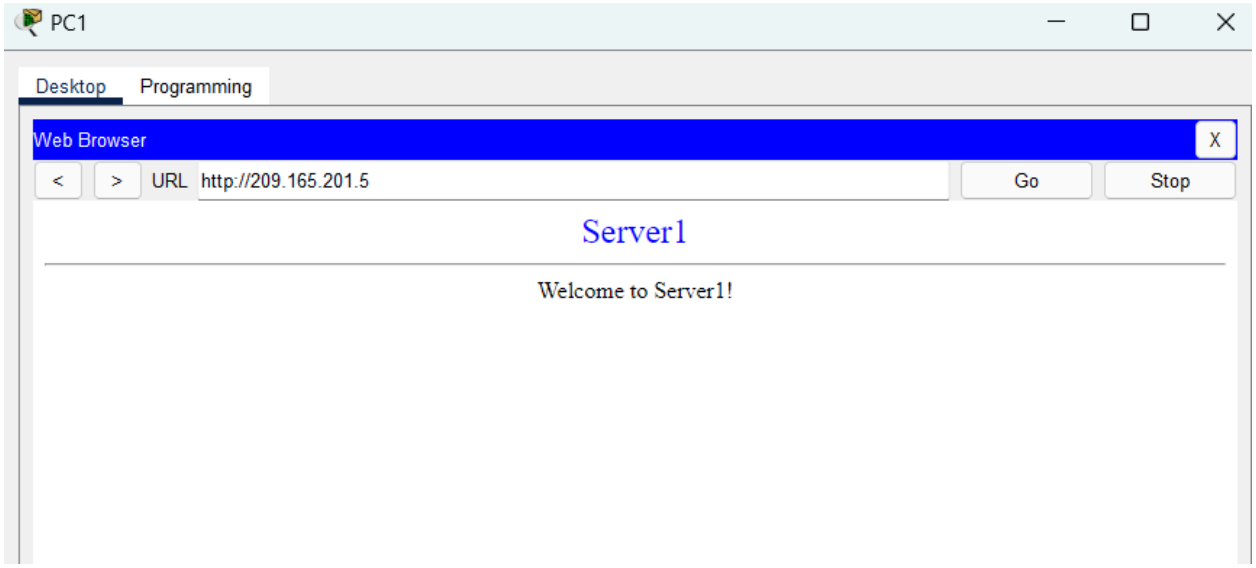


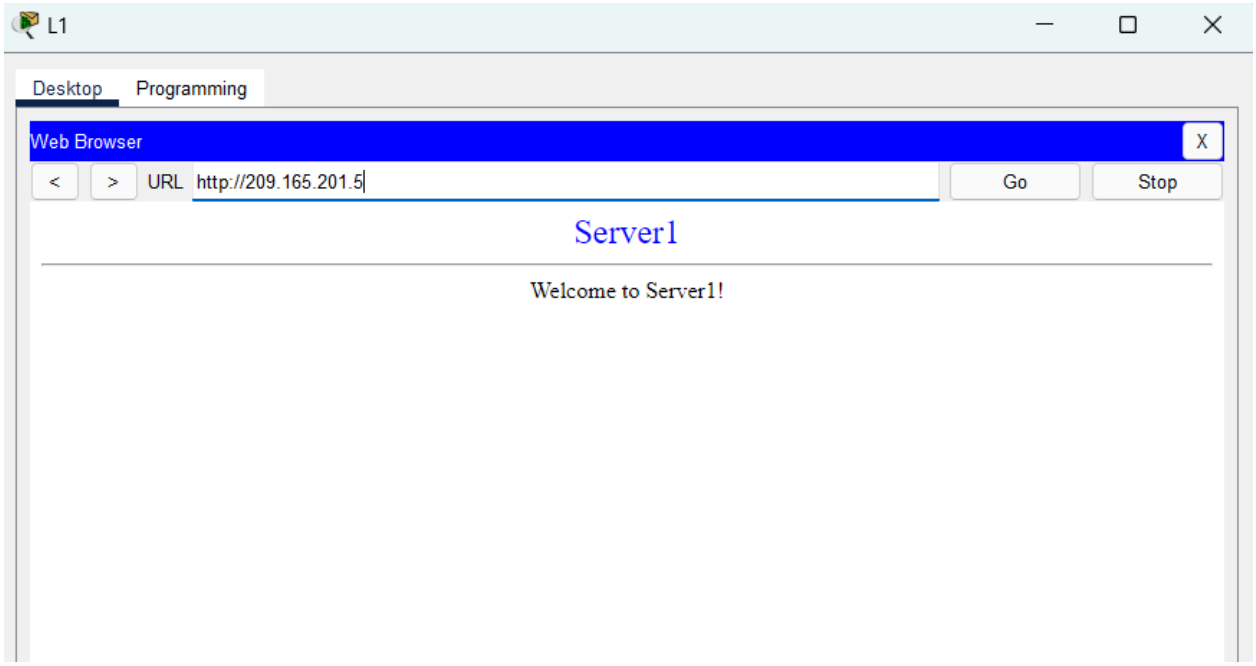
Working Procedure:

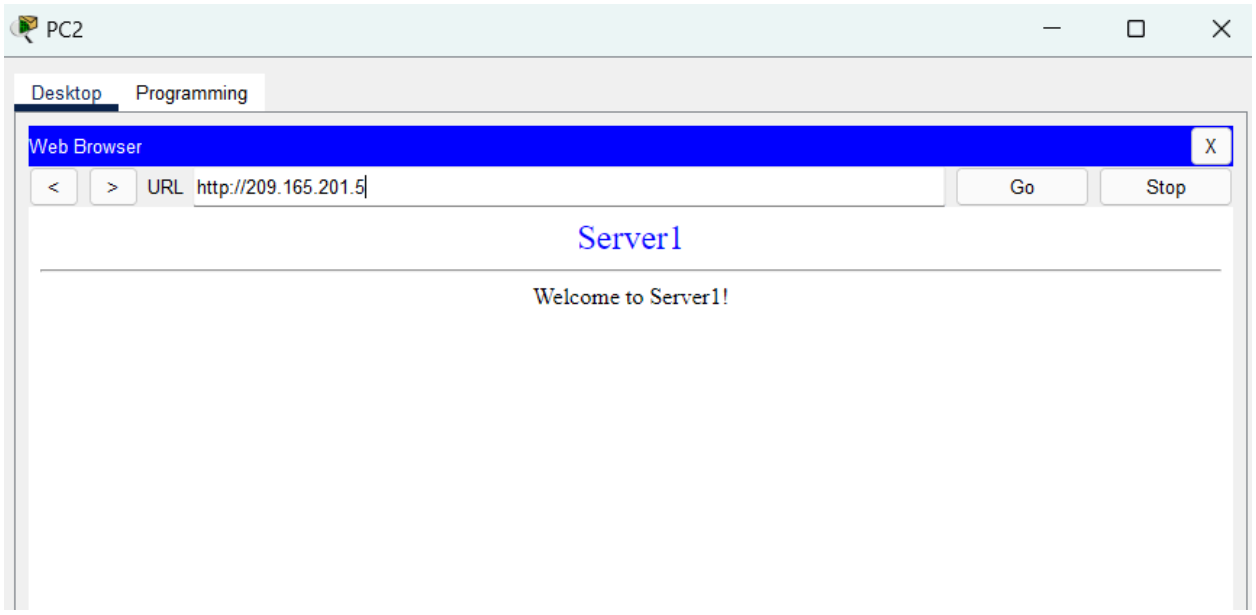

R1
CLI
IOS Command Line Interface
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
R1>
R1>
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#ac
R1(config)#access-list 1 permit 172.16.0.0 0.0.255.255
R1(config)#
R1(config)#ip nat pool ANY_POOL_NAME 209.165.200.233 209.165.200.234 netmask 255.255.255.252
R1(config)#ip nat inside source list 1 pool ANY_POOL_NAME overload
R1(config)#
R1(config)#int
R1(config)#interface s0/1/0
R1(config-if)#ip nat outside
R1(config-if)#interface g0/0/0
R1(config-if)#ip nat inside
R1(config-if)#interface g0/0/1
R1(config-if)#ip nat inside
R1(config-if)#
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#
R1#show ip nat translations
Pro Inside global Inside local Outside local Outside global
tcp 209.165.200.233:1024172.16.11.10:1025 209.165.201.5:80 209.165.201.5:80
tcp 209.165.200.233:1025172.16.10.10:1025 209.165.201.5:80 209.165.201.5:80
tcp 209.165.200.233:1026172.16.11.10:1026 209.165.201.5:80 209.165.201.5:80
R1#
Copy
Paste

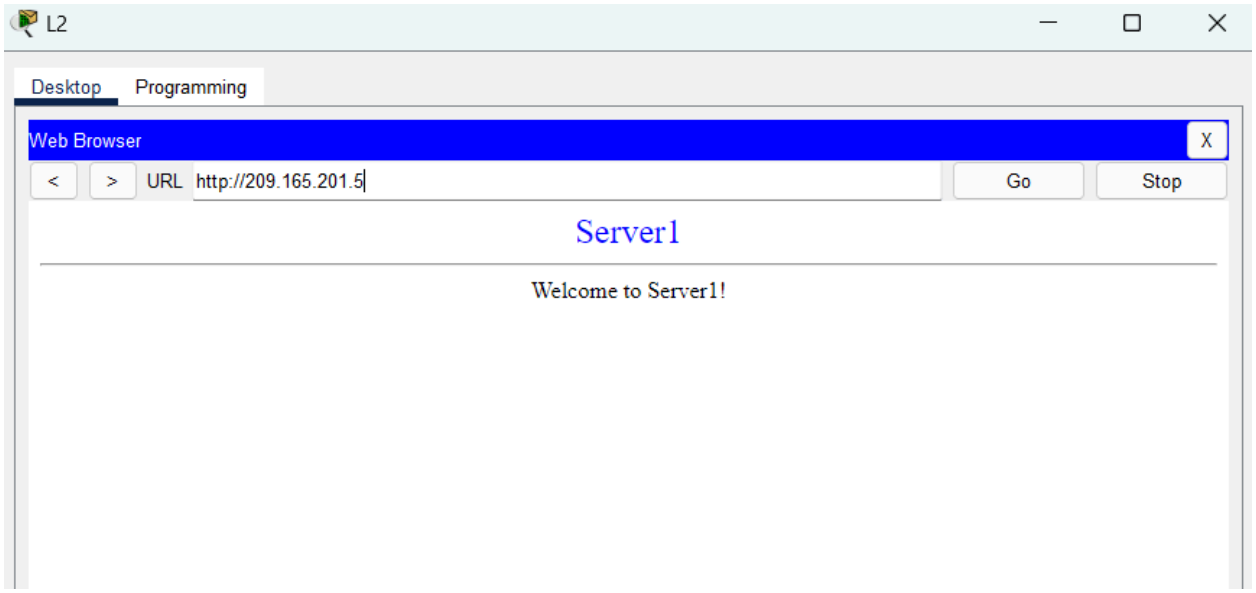

R2
CLI
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
R2>
R2>
R2>
R2>
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#ac
R2(config)#access-list 2 permit 172.17.0.0 0.0.255.255
R2(config)#
R2(config)#ip nat inside source list 2 interface s0/1/1 overload
R2(config)#
R2(config)#interface s0/1/1
R2(config-if)#ip nat outside
R2(config-if)#interface g0/0/0
R2(config-if)#ip nat inside
R2(config-if)#interface g0/0/1
R2(config-if)#ip nat inside
R2(config-if)#
R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
R2#
R2#copy run
R2#copy running-config st
R2#copy running-config startup-config
R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
R2#
R2#
R2#
R2#
Copy
Paste

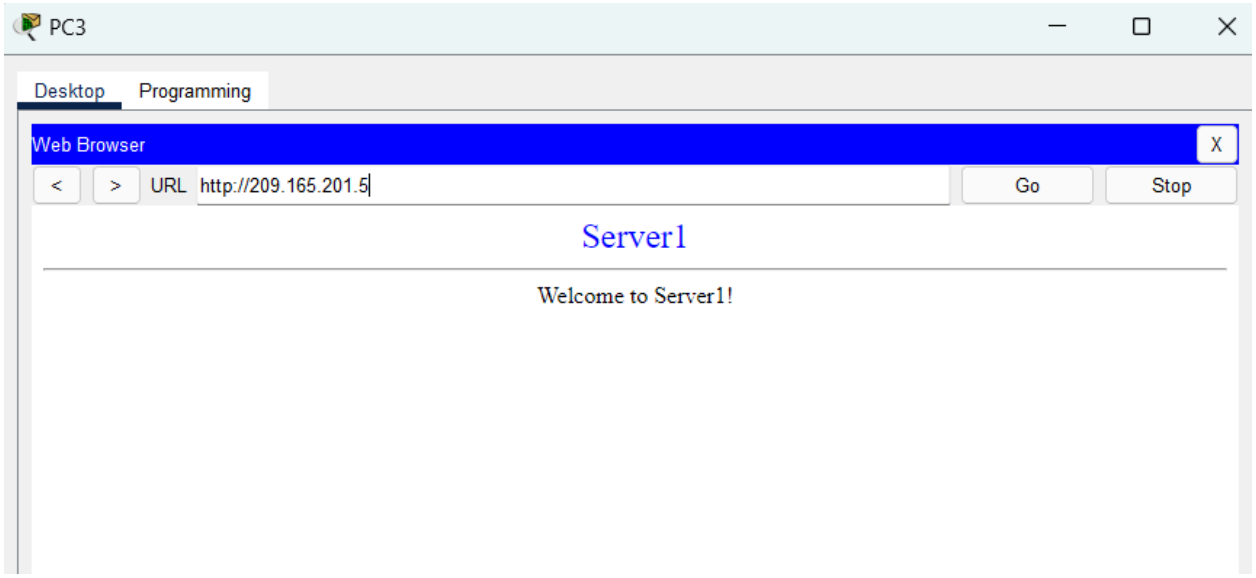
Observation:

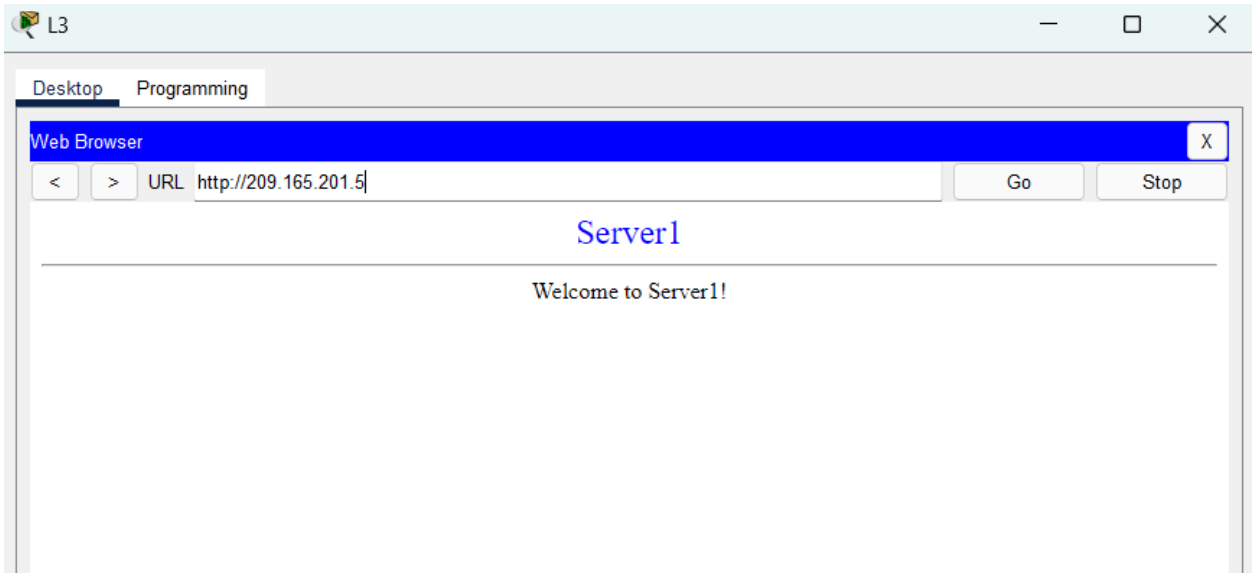


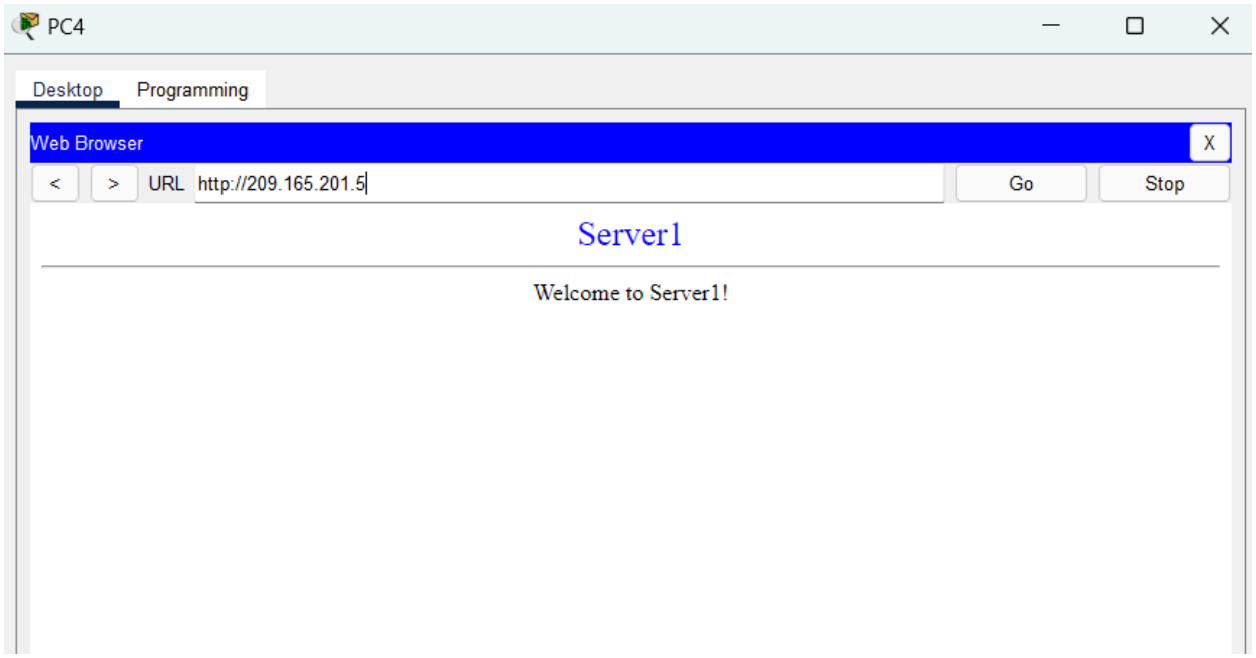


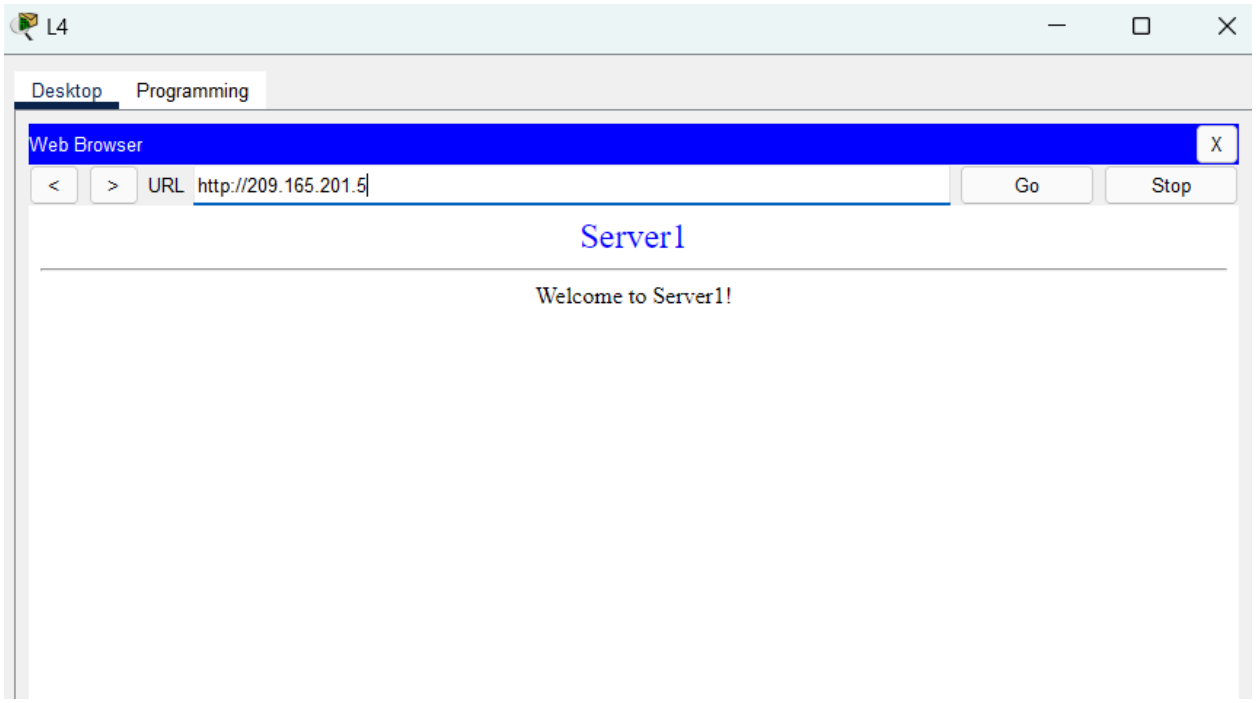

PC2
Desktop
Programming
Web Browser
X
<
>
URL
http://209.165.201.5
Go
Stop
Server1
Welcome to Server1!


L2
Desktop
Programming
Web Browser
X
<
>
URL
http://209.165.201.5
Go
Stop
Server1
Welcome to Server1!


PC3
Desktop
Programming
Web Browser
X
<
>
URL
http://209.165.201.5
Go
Stop
Server1
Welcome to Server1!


L3
Desktop
Programming
Web Browser
X
<
>
URL
http://209.165.201.5
Go
Stop
Server1
Welcome to Server1!





Task # 02:

1. From the web browser of each of the PCs that use R1 as their gateway (PC1, L1, PC2, and L2), access the web page for Server1.
   Question:
   Were all connections successful?
   Ans: Yes. The screenshots prove that.

2. From the web browser of each of the PCs that use R2 as their gateway (PC3, L3, PC4, and L4), access the web page for Server1.
   Question:
   Were all connections successful?
   Ans: Yes. The screenshots prove that.

3. Compare the NAT statistics on the two devices.
   Question:
   Why doesn't R2 list any dynamic mappings?
   Ans: In R2, a pool of global IP addresses were not configured which is why R2 does not list any dynamic mappings unlike R1.